

torch.allclose#

<https://pytorch.org/docs/stable/generated/torch.allclose.html>

Description:

This function checks if input and other satisfy the condition: elementwise, for all elements of input and other. The behaviour of this function is analogous to `numpy.allclose` input (Tensor) – first tensor to compare other (Tensor) – second tensor to compare atol (float, optional) – absolute tolerance. Default: 1e-08

Function Signature

```
torch.allclose(input: Tensor, other: Tensor, rtol: float = 1e-05, atol: float = 1e-08, equal_nan: bool = False) → bool#
```

Parameters

Code Examples

```
>>> torch.allclose(torch.tensor([10000., 1e-07]),
torch.tensor([10000.1, 1e-08]))
False
>>> torch.allclose(torch.tensor([10000., 1e-08]),
torch.tensor([10000.1, 1e-09]))
True
>>> torch.allclose(torch.tensor([1.0, float('nan')]),
torch.tensor([1.0, float('nan')]))
False
>>> torch.allclose(torch.tensor([1.0, float('nan')]),
torch.tensor([1.0, float('nan')]), equal_nan=True)
True
```

Detailed Documentation

Documentation

This function checks if input and other satisfy the condition:

elementwise, for all elements of input and other. The behaviour of this function is analogous to `numpy.allclose`

input (Tensor) – first tensor to compare

other (Tensor) – second tensor to compare

atol (float, optional) – absolute tolerance. Default: 1e-08

rtol (float, optional) – relative tolerance. Default: 1e-05

equal_nan (bool, optional) – if True, then two NaN s will be considered equal. Default: False

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